David Rimmer

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# PHAS2000

# Level 2 Coding Project Report

# 1.1: 10keV data

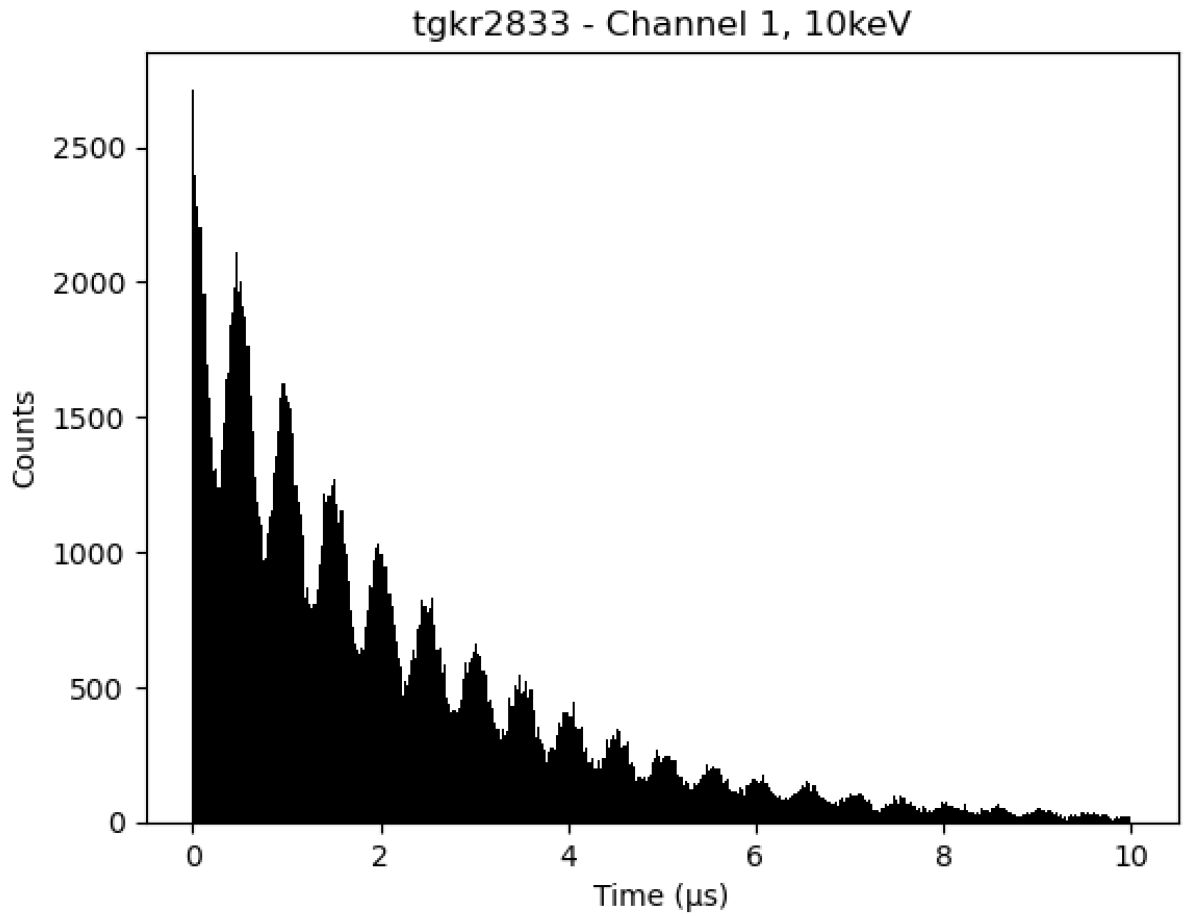



**Figure 1: Histogram plot of the left channel for counts against time for 10keV with a bin number of the square root of the length of the data.**

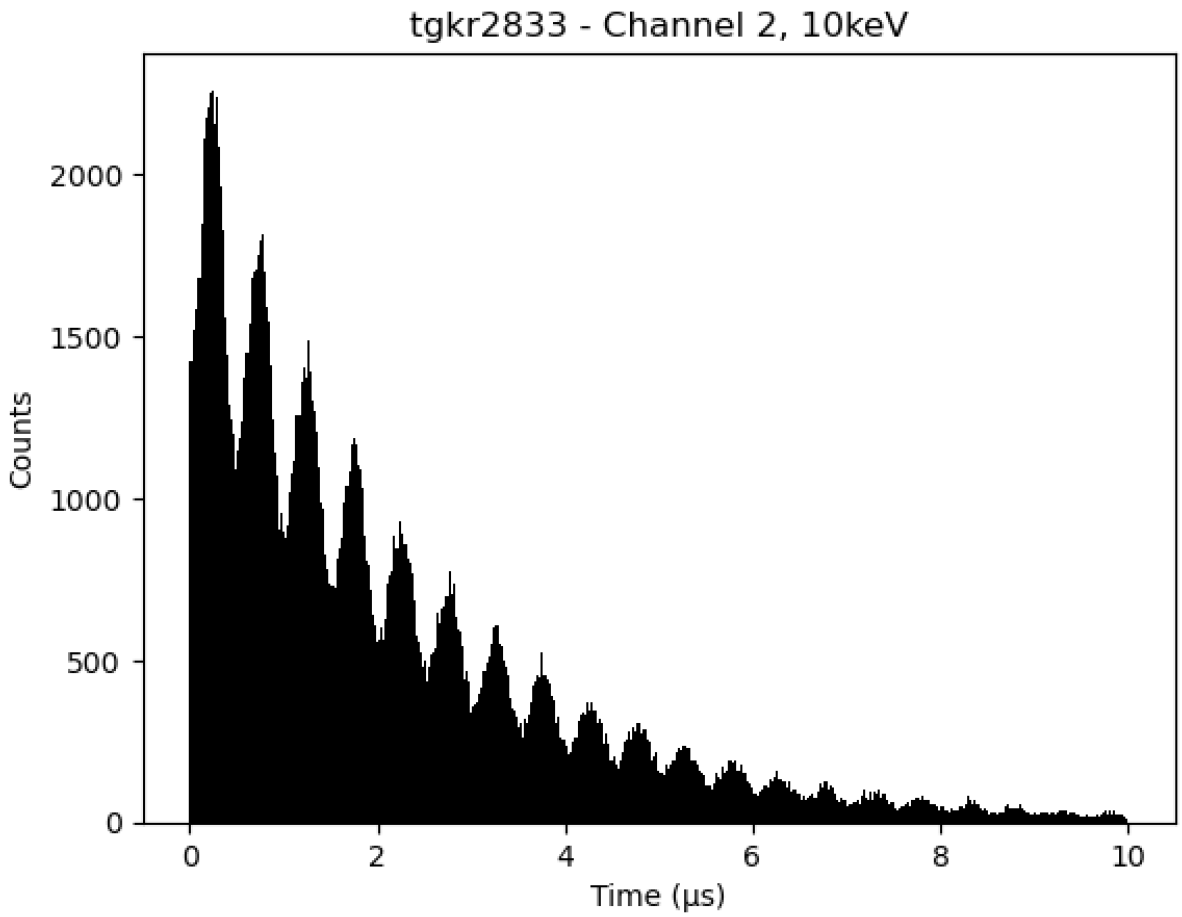



**Figure 2: Histogram plot of the Right channel for counts against time for 10keV with a bin number of the square root of the length of the data.**

# 1.2: Asymmetry plot

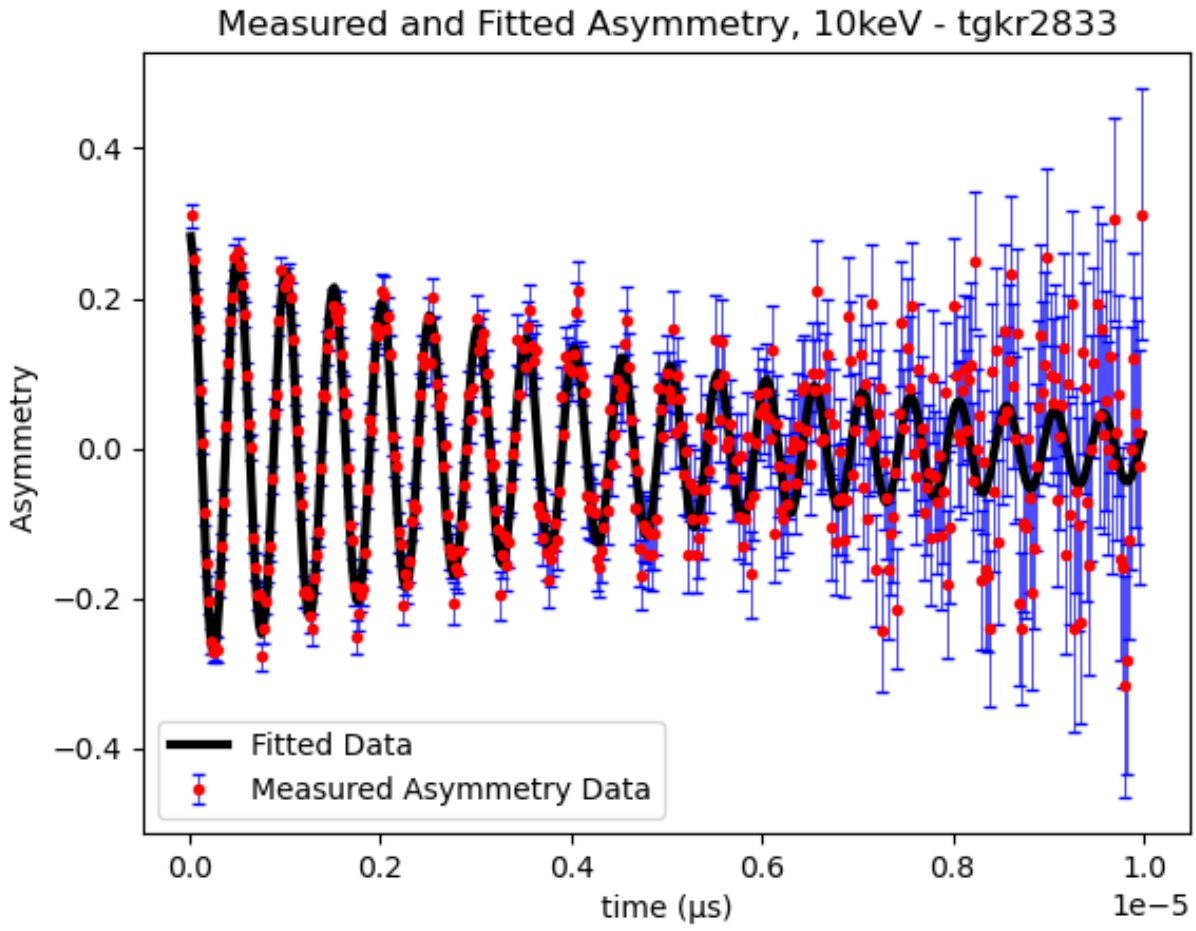



**Figure 3: Plot of the measured asymmetry data and fitted asymmetry data with B = $0.014642 \pm 8 \times 10^{-6}$ (Tesla), Beta = $0.94 \pm 0.04$ (Radians) and Tau = $5.3 \times 10^{-6} \pm 3 \times 10^{-7}$ (seconds).**

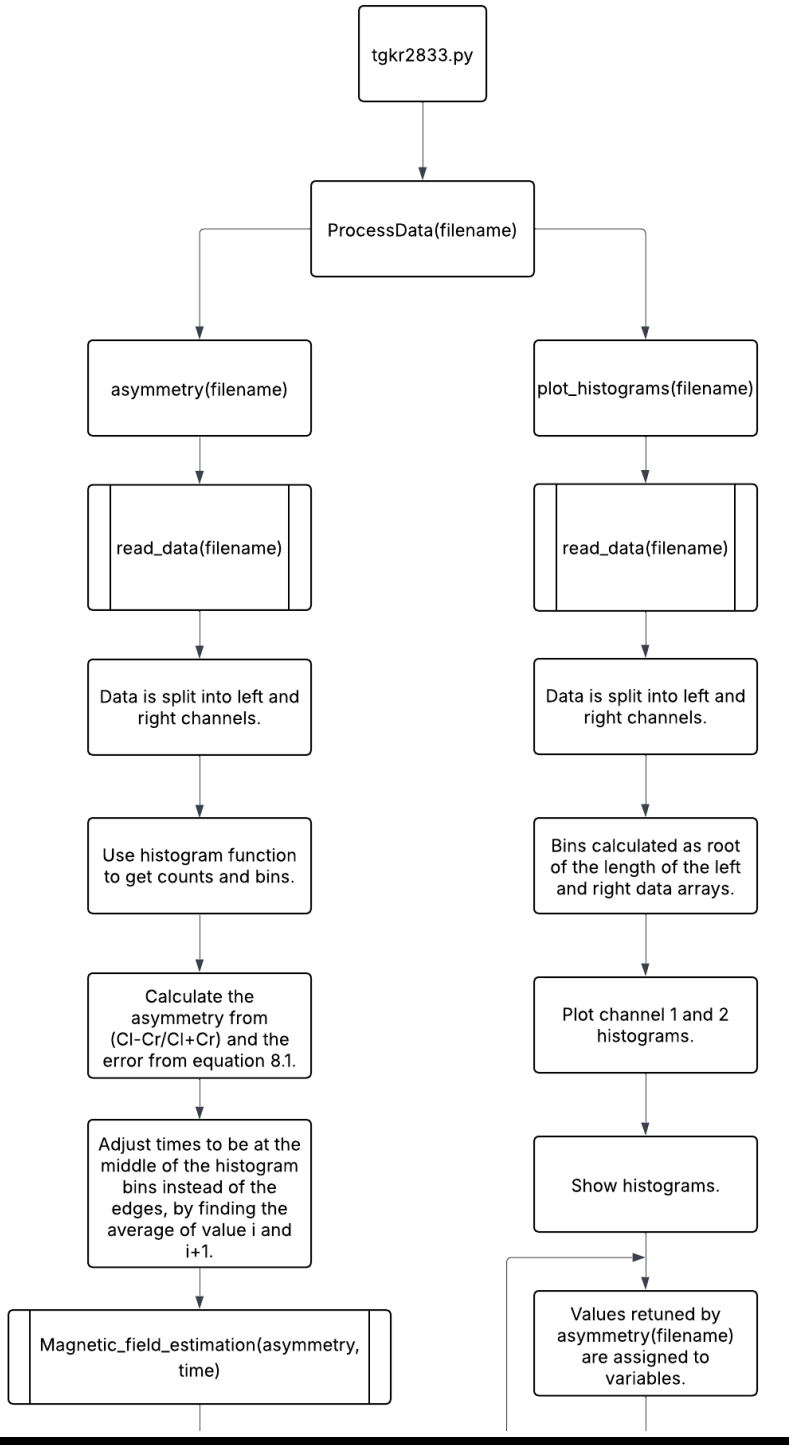
# 1.3: Table of Results

**Table 1: Results of processing the assessment data file**

| | |
|---|---|
| **Beta, $\beta$ (Radians):** | $0.94 \pm 0.04$ |
| **Magnetic Field, B (T, Tesla):** | $0.014642 \pm 8 \times 10^{-6}$ |
| **Time Damping, $\tau$ (s, Seconds):** | $5.3 \times 10^{-6} \pm 3 \times 10^{-7}$ |

# 2: Flowcharts

# 2.1: ProcessData()



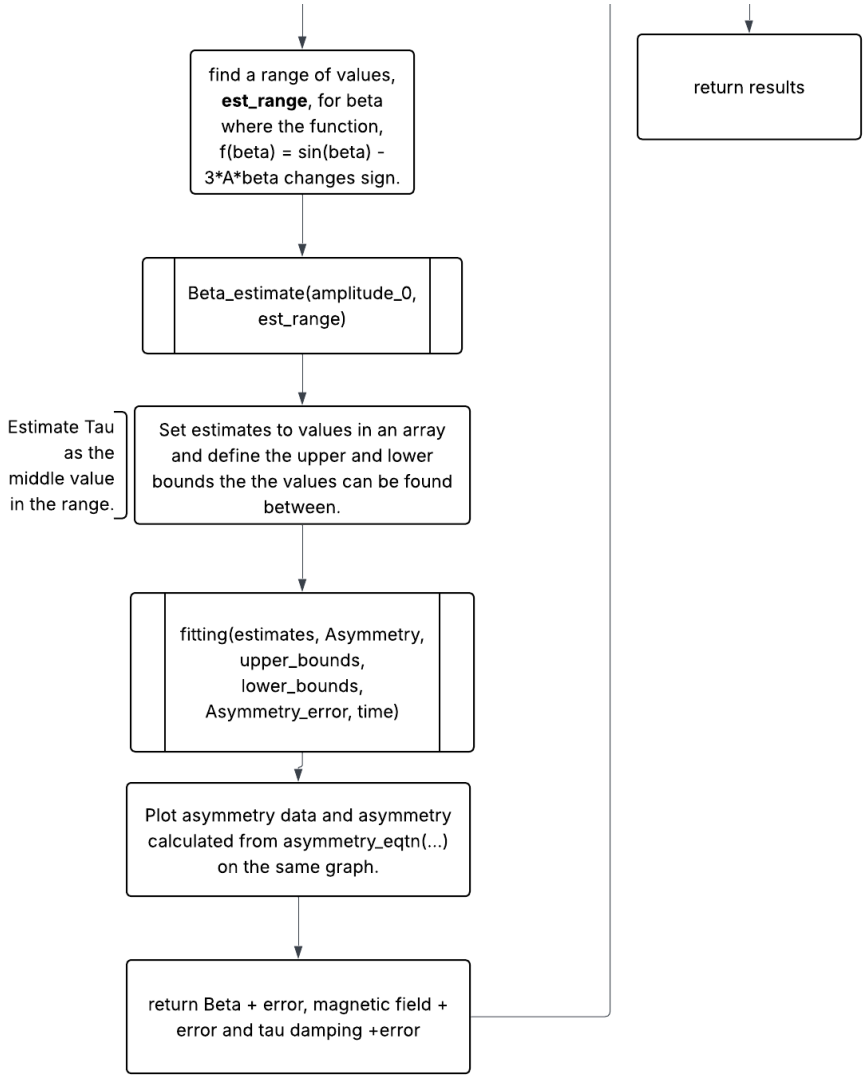



**Figure 4: Flowchart of the function ProcessData()**

# 2.2: read_data()

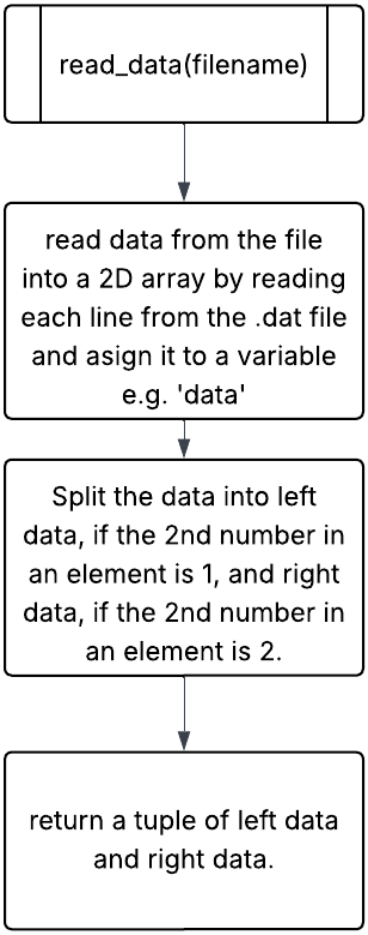



**Figure 5: Flowchart of read_data()**

# 2.3: plot_histograms()

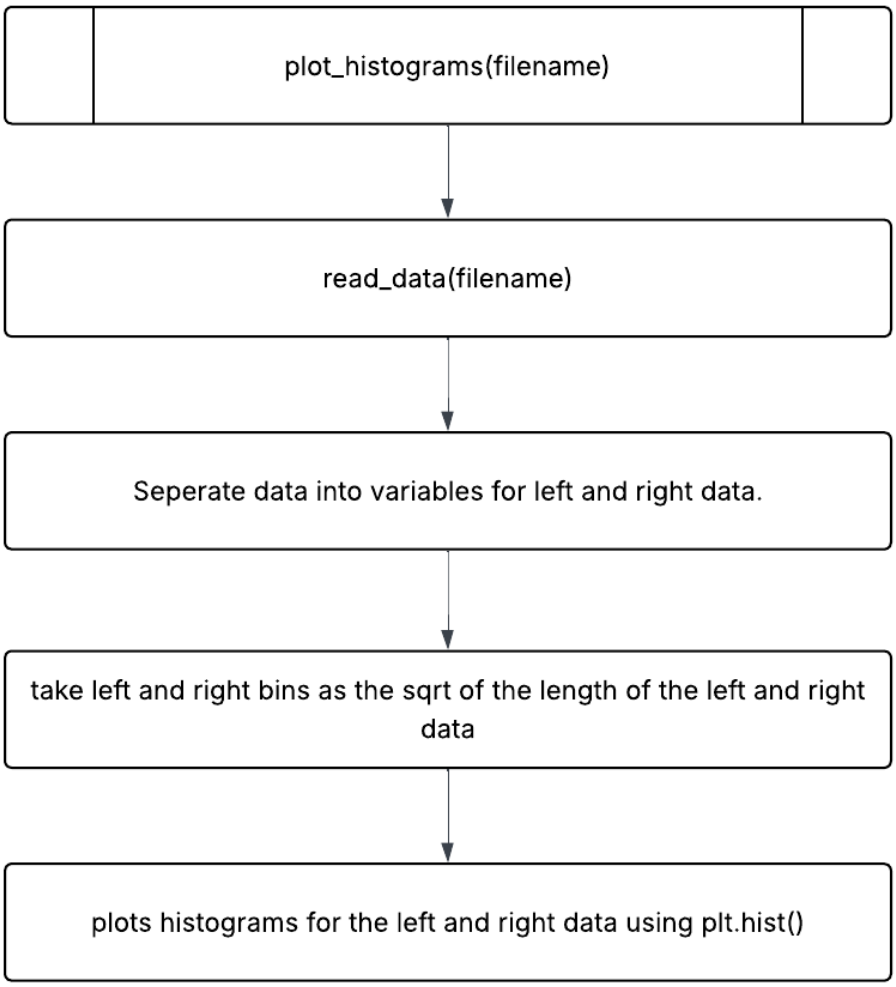



**Figure 6: Flowchart of plot_histograms()**

# 2.4: Magnetic_Field_Estimation()

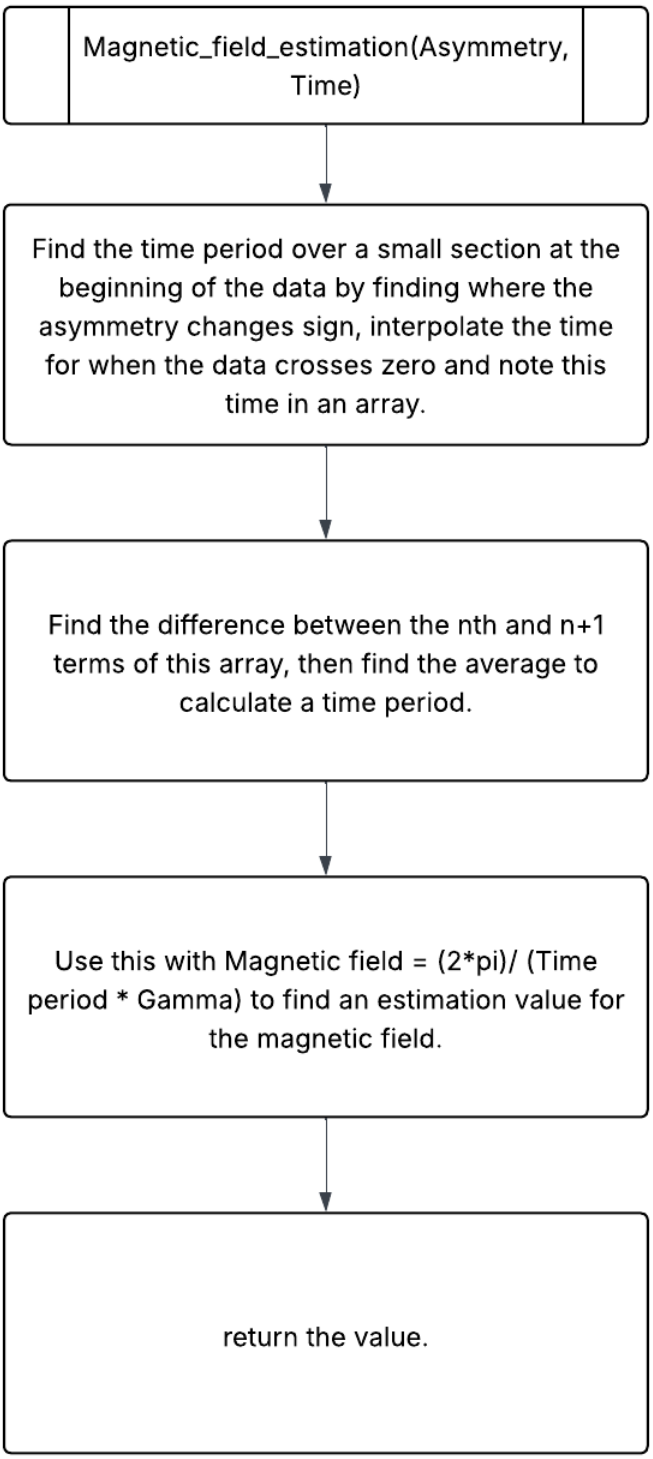



**Figure 7: Flowchart of Magnetic_Field_Estimation()**

# 2.5: asymmetry_equation()

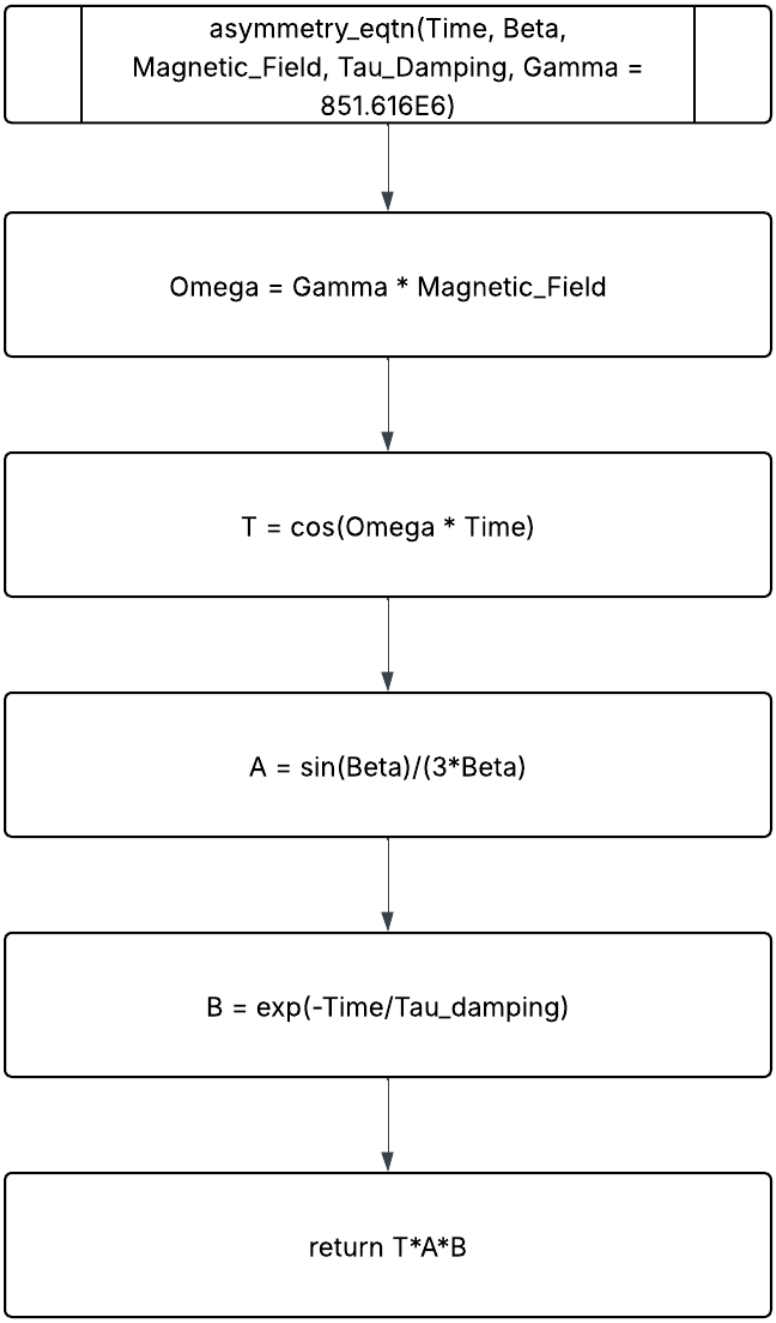



**Figure 8: Flowchart of asymmetry_equation()**

# 2.6: fitting()

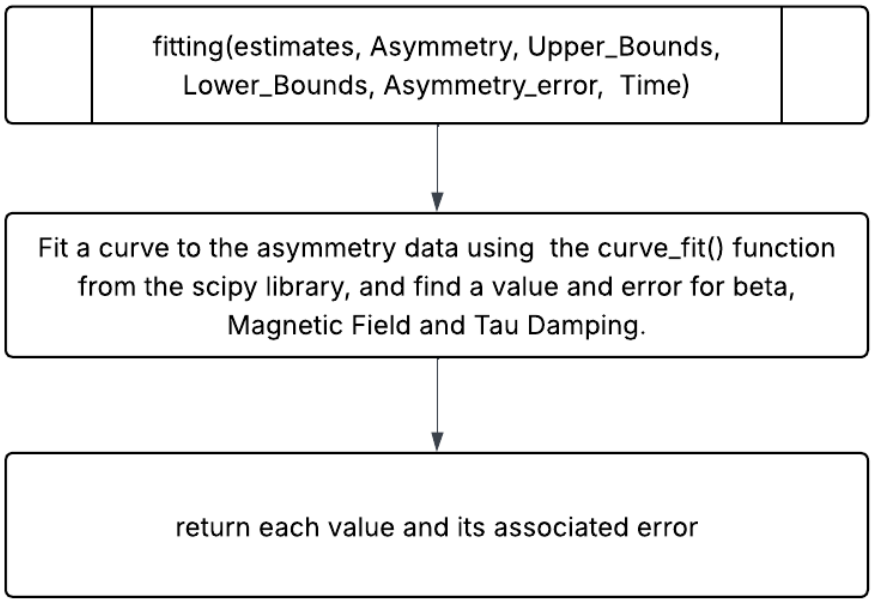



**Figure 9: Flowchart of fitting()**

# 2.7: Amplitude_eqtn()

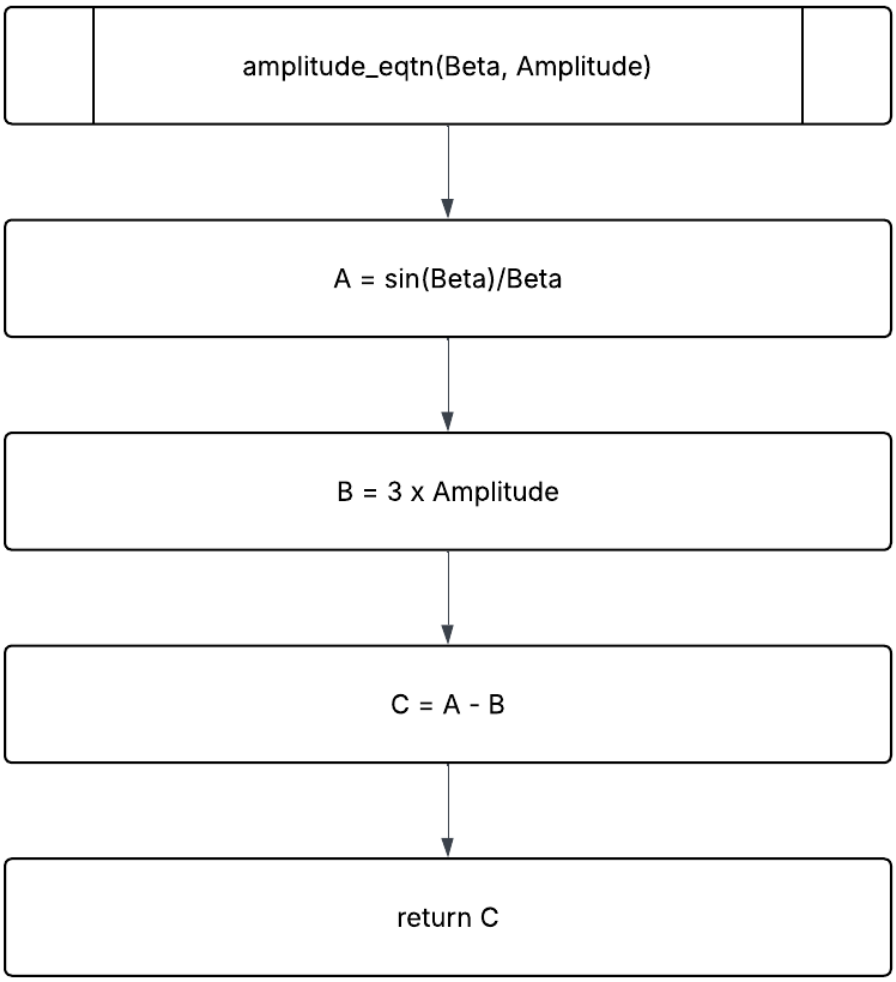



**Figure 10: Flowchart of Amplitude_eqtn()**

# 2.8: Beta_estimate()

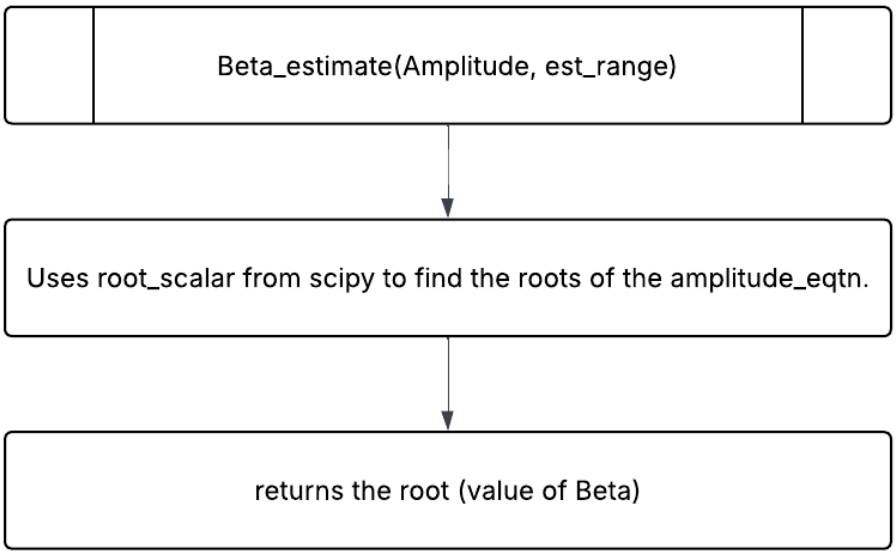



**Figure 11: Flowchart of Beta_estimate()**